DOCUMENTATION

Introduction

The rapid spread of online news has increased the circulation of misinformation, making it difficult for readers and organizations to verify content quickly. Traditional manual fact-checking processes are time-consuming and allow fake news to spread before validation. This project uses Natural Language Processing (NLP) to automatically classify news articles as fake or real, enabling faster verification, improving trust in credible sources, and reducing the impact of false information.

Problem Statement

Develop an automated NLP-based classification system that identifies whether a news article is fake or real within seconds, reducing manual review time from multiple days to near real-time, while achieving an F1-score of at least 85% on a labeled news dataset.

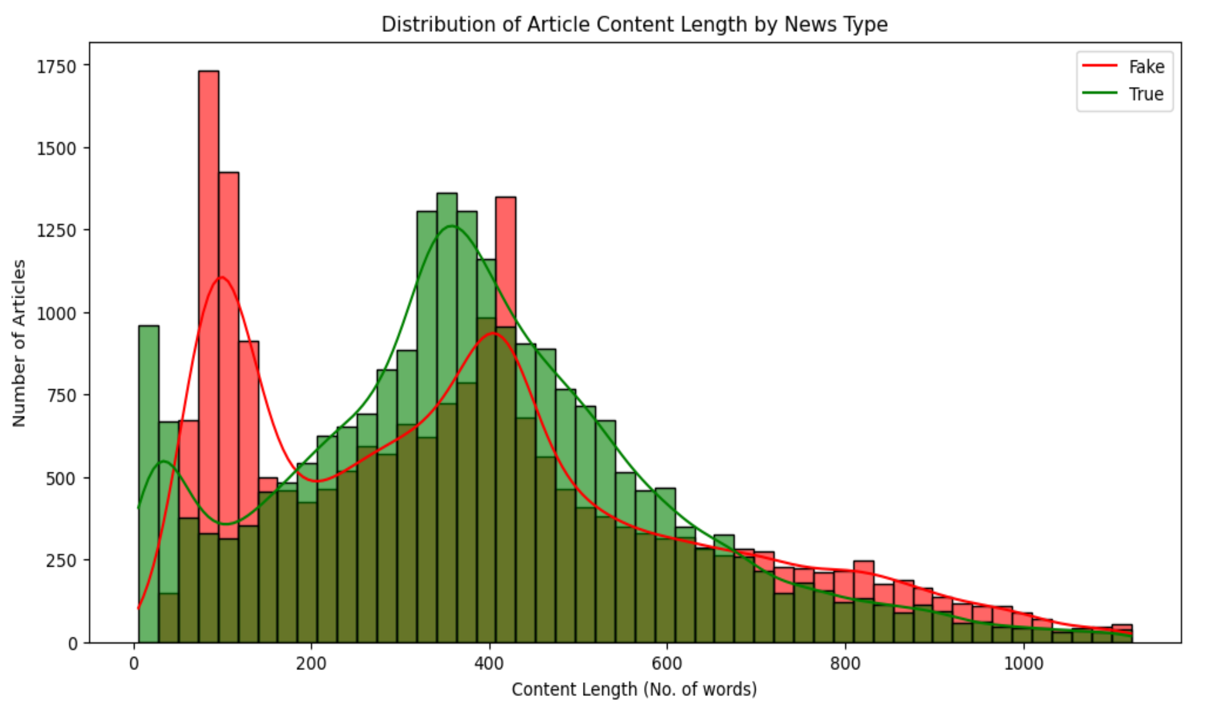
Data Wrangling

The dataset was first created by combining True and Fake news sources, with a new column status added to label True news as 1 and Fake news as 0. There were total 44898 rows and 5 columns for the dataset at first. The title and text columns were combined into a single content column to serve as the main text for analysis. The date column was cleaned. The text data was cleaned by converting all text to lowercase, removing punctuation and special characters, and eliminating common English stopwords. For natural language processing, the text was tokenized into individual words and then lemmatized using WordNet to reduce words to their base forms, resulting in lemmatized_tokens, which were later recombined into lemmatized_content for modeling purposes. Additionally, a numeric feature content_length was created to capture the length of each article.

To investigate whether fake news articles exhibit more aggressive or toxic language, toxicity scores were generated using the Detoxify model. Six toxicity-related features were extracted, including toxicity, severe toxicity, obscene language, threats, insults, and identity attacks. The dataset is now fully cleaned, structured, and prepared for exploratory data analysis, frequency analysis and subsequent machine learning modeling, with optional further steps like removing domain-specific high-frequency words for more targeted analysis. After all these cleaning process, we have now 43556 rows and 16 columns in the data frame.

Exploratory Data Analysis

During EDA, the dataset showed a fairly balanced distribution between Fake (≈52%) and True (≈48%) news. Text characteristics such as content length and token counts indicated that shorter articles are slightly more likely to be fake, though not strongly predictive.



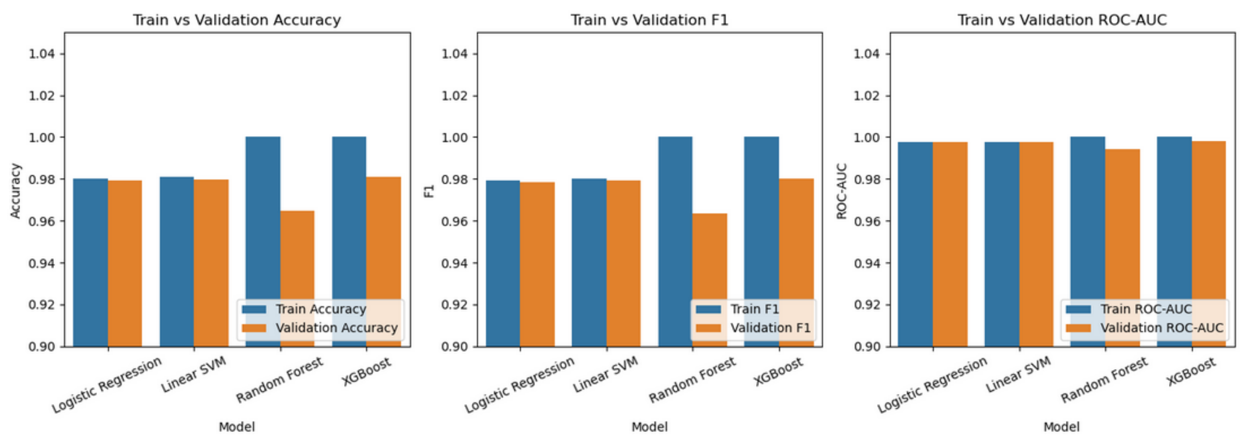
Fake news often contained sensational or politically charged terms, while True news used formal reporting words. Toxicity analysis using Detoxify revealed that Fake news consistently exhibits higher levels of aggressive or offensive language across features like toxicity, insult, obscene, and identity_attack. These insights into word usage, categories, and toxicity help differentiate Fake vs True news and inform preprocessing for modeling.

Model Preprocessing with Feature Engineering

Average word length and unique word ratio were engineered to capture the structural properties of news articles, while features likely to introduce bias, such as the subject category, were removed. The dataset was split into stratified training and testing sets to preserve class balance. Numeric features were standardized, and textual data was transformed into dense vector representations using Word2Vec embeddings trained exclusively on the training data to prevent data leakage, resulting in a final feature set suitable for downstream modeling.

Algorithms used to build the model with evaluation metrics

Several machine learning models, including Logistic Regression, Linear SVM, Random Forest, and XGBoost, were trained and evaluated using stratified cross-validation with metrics such as accuracy, precision, recall, F1-score, and ROC-AUC. Logistic Regression and Linear SVM achieved the strongest and most consistent performance, with F1-scores close to 0.98 and ROC-AUC values above 0.99.

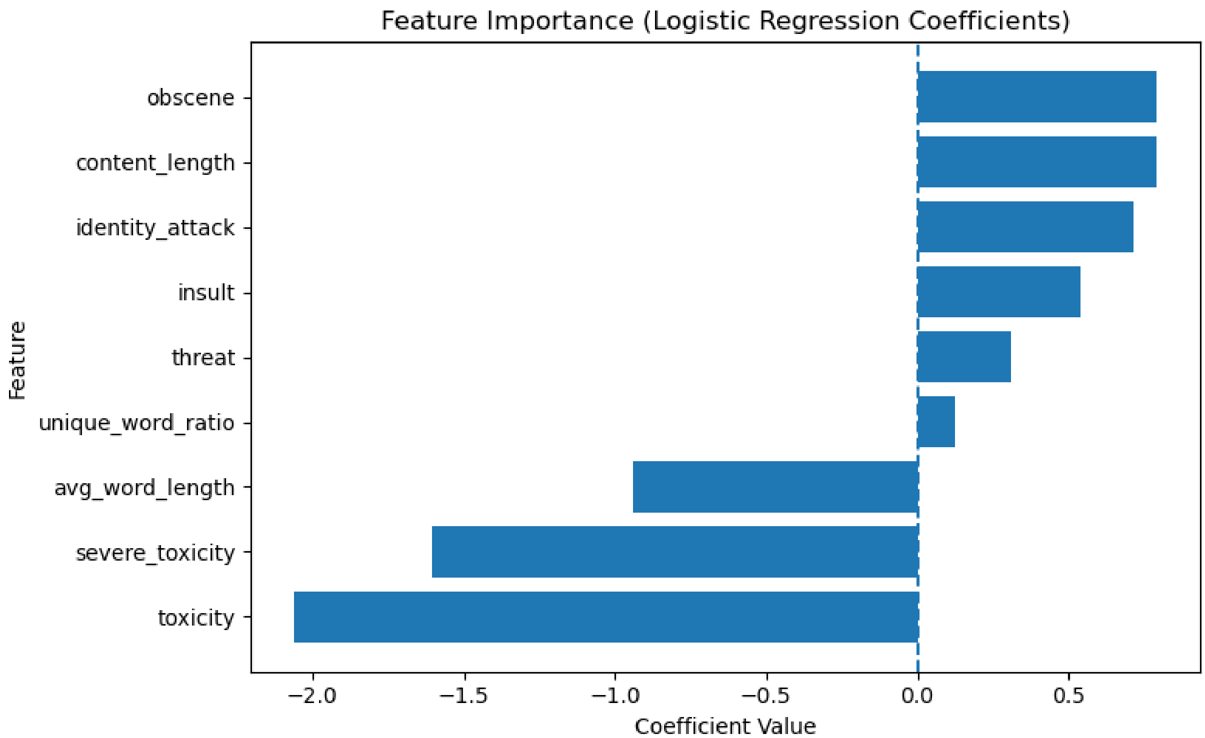


Hyperparameter tuning was applied to the top models using RandomizedSearchCV, after which Logistic Regression was selected as the final model due to its comparable performance, faster training time, probabilistic outputs, and interpretability.

Winning Model

After hyperparameter tuning, both Logistic Regression and Linear SVM achieved strong and comparable performance, with F1-scores close to 0.98 and ROC-AUC values above 0.99. Although Linear SVM achieved slightly higher recall for the minority class, Logistic Regression was selected as the final model due to its faster training time, ability to provide probabilistic outputs, and better scalability for large datasets, while maintaining nearly identical predictive performance.

Feature importance analysis showed that content length, lexical diversity, and some toxicity-related features (obscene, threat, insult, identity attack) were positively associated with true news, while average word length and other toxicity scores (toxicity, severe toxicity) were negatively associated. The presence of some toxicity-related terms in true news may reflect the reporting of controversial or sensitive topics in legitimate news coverage. Higher general toxicity scores and severe toxic language are more associated with fake news, suggesting that fake news articles may contain more aggressive or emotionally charged language.



Recommendations

The developed fake news classification system demonstrates strong predictive performance and can be used as a supporting tool for journalists, fact-checkers, and online platforms to identify potentially misleading content. News organizations and social media platforms can integrate such models into their moderation pipelines to flag suspicious articles for further verification. Additionally, combining linguistic features with contextual signals such as source credibility or network propagation patterns could further enhance detection accuracy.

Limitations

Despite achieving strong performance, the model has several limitations. The dataset used for training may not fully represent the diversity of real-world news articles, which could affect the model's ability to generalize to unseen sources or writing styles. The toxicity analysis was performed only on the first portion of each article to reduce computational cost, which may overlook toxic language appearing later in the text. Additionally, Word2Vec embeddings capture semantic relationships but may not fully capture contextual meaning compared to more advanced transformer-based models.

Conclusion

This project successfully developed a machine learning-based system for detecting fake news using natural language processing techniques. By combining text embeddings, linguistic features, and toxicity analysis, multiple classification models were evaluated, with Logistic Regression emerging as the best-performing model. The final model achieved strong predictive performance with F1-scores near 0.98 and ROC-AUC values above 0.99. These results demonstrate the effectiveness of combining traditional NLP techniques with engineered features for fake news detection.

Future Scope of Work

Future work could explore more advanced language models such as BERT or other transformer-based architectures to capture deeper contextual relationships within news articles. Incorporating additional data sources, such as publisher credibility, social media engagement patterns, or article metadata, may further improve detection accuracy. Deploying the model as a real-time API or web application would also allow users to instantly evaluate the credibility of new articles.